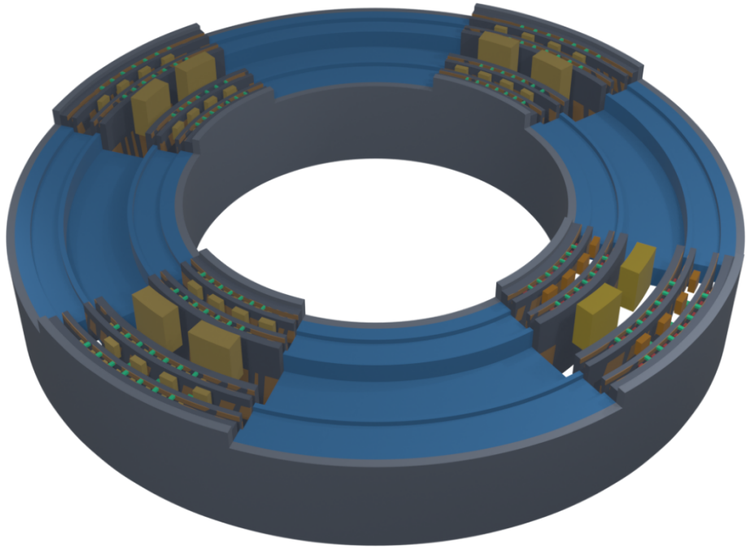
**Unified 3D Fractal Switch (H3-Cantor-Robust, b = 3, g_min = 27 nm) vs. Five Iso-Area Matched Controls**

**Sector 0 Key: Gold = Gen-1 (W=810 nm, z=-360 nm) | Amber = Gen-2 (W=270 nm, z=-120 nm) | Crimson = Gen-3 (W=90 nm, z=-40 nm) | Emerald = Rim Micro-Bridges | Sapphire = Landing Pads**

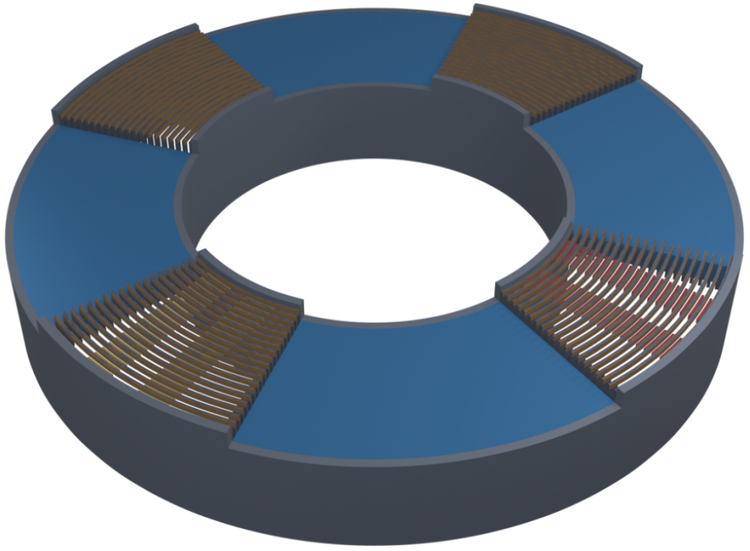
### (a) H3-Cantor-Robust

Tracks=7/sec | Lobes=42/sec | Bridges=35/sec | g_min=27 nm
A_tooth=3.1026 um² | Fz(3°)=+2.47 pN | Bz=2.28 | Y_lab=85.9% | PASS



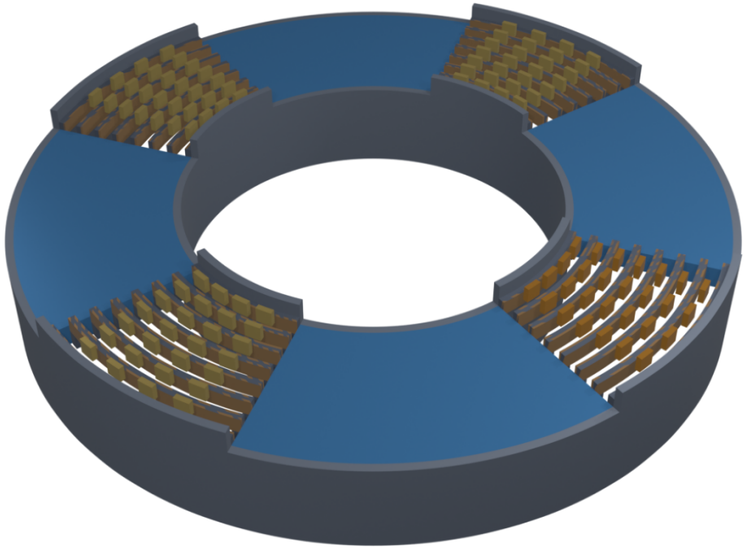
### (b) Ctrl-KnifeEdge-g27

Tracks=19/sec | Lobes=38/sec | Bridges=0/sec | g_min=27 nm
A_tooth=3.1026 um² | Fz(3°)=+2.20 pN | Bz=1.28 | Y_lab=41.5% | PASS



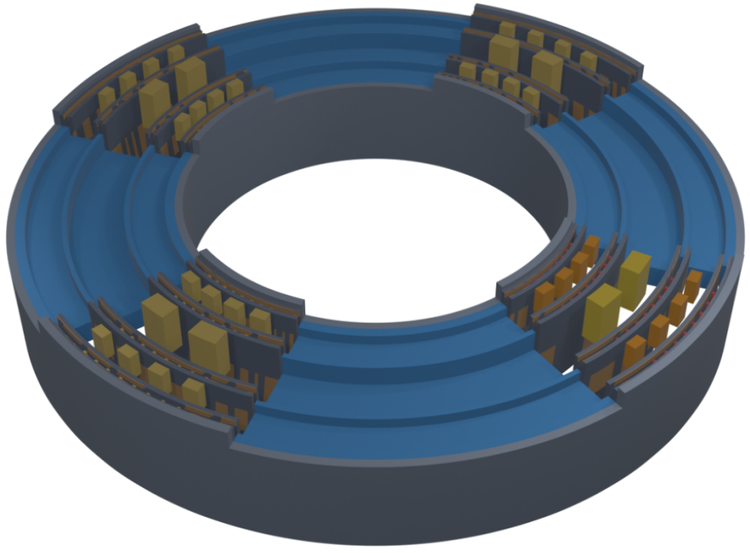
### (c) Ctrl-Uni-7Trk

Tracks=7/sec | Lobes=28/sec | Bridges=0/sec | g_min=73 nm
A_tooth=3.1026 um² | Fz(3°)=-1.57 pN | Bz=1.47 | Y_lab=1.1% | PASS



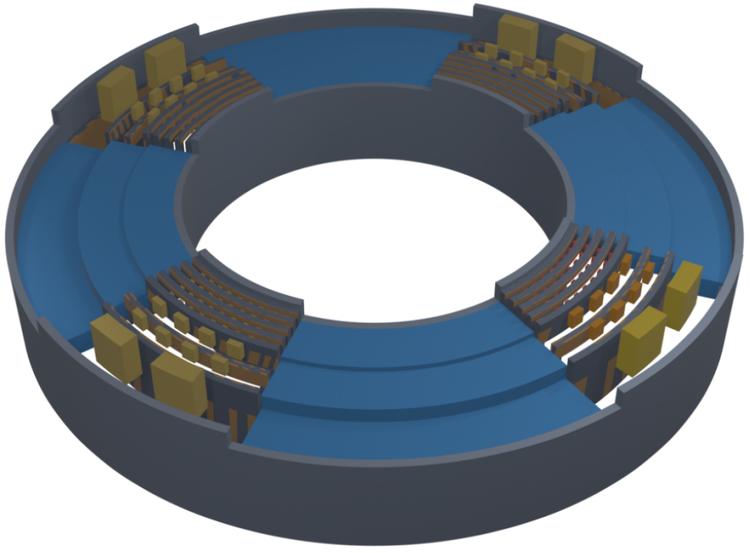
### (d) Ctrl-Lin-3Stage

Tracks=7/sec | Lobes=42/sec | Bridges=0/sec | g_min=27 nm
A_tooth=3.1026 um² | Fz(3°)=+1.90 pN | Bz=2.53 | Y_lab=88.0% | PASS



### (e) Ctrl-Mono-3Stage

Tracks=7/sec | Lobes=42/sec | Bridges=0/sec | g_min=27 nm
A_tooth=3.1026 um² | Fz(3°)=+6.19 pN | Bz=2.56 | Y_lab=89.0% | PASS



### (f) Ctrl-Shuf-OutPhase

Tracks=7/sec | Lobes=24/sec | Bridges=0/sec | g_min=27 nm
A_tooth=3.1026 um² | Fz(3°)=-1.10 pN | Bz=0.48 | Y_lab=0.1% | PASS

